STEP 1 TOWARD O1*: THE FLUCTUATION OPERATOR CARRIES THE PAINLEVÉ II SIGNATURE, AND ITS a-FLOW IS ISOMONODROMIC

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Abstract. Continuation of the O1 note. O1* asks that the weak-noise fluctuation coefficients $c_n(a)$ of the TW_β tail be the trans-series coefficients of Painlevé II, so that Costin's Borel-summability theorem applies. This note carries out the second variation of the Freidlin–Wentzell action about the Airy instanton and finds the fluctuation (Jacobi) operator

$$\mathcal{J} = -\partial_x^2 + \big(6p_\star^2 - 2(x+a)\big),$$

whose $6p_\star^2$ term is exactly the coefficient of the Painlevé II linearization $-\partial_s^2+(6q^2+s)$ — a nontrivial fingerprint, not a coincidence of normalization. It then frames the identification correctly: the fixed-level linear problem is Airy (trivial deformation), and the Painlevé II structure lives in the *spectral-level* variable a, where the Tracy–Widom theorem makes the $\beta = 2$ tail *exactly* the Hastings–McLeod τ-function. Step 1 thereby reduces to one precise obligation — identify the fluctuation determinant / transport hierarchy with the Jimbo–Miwa τ-function of the a-isomonodromic Painlevé II family — for which the machinery (Flaschka–Newell linear problem, JMU deformation theory) is classical and Costin then closes summability.

1. Setup and the goal of Step 1

From the O1 note: the tail $\mathcal{T}_\beta(a) = \mathbb{P}(\text{Riccati explosion})$ has the weak-noise (large-β, $\varepsilon^2 = 1/\beta$) form $\mathcal{T}_\beta(a) = e^{-\beta\Phi(a)}\sum_{n\ge 0} c_n(a)\,\beta^{-n}$, where the eikonal instanton $p_\star(x)$ solves the fluctuation-branch Riccati equation

$$\dot{p_\star} = p_\star^2 - (x+a), \qquad p_\star = -\varphi'/\varphi, \quad \varphi'' = (x+a)\varphi \ \ (\text{Airy}), \tag{1}$$

and $\Phi(a) = \frac{1}{2}\int b^2\,\mathrm{d}x$ with $b := (x+a) - p_\star^2 = -\dot{p_\star}$. The c_n are generated by the transport hierarchy of the fluctuation operator $\mathcal{J} := \frac{1}{2}\delta^2 I[p_\star]$, the Hessian of the Freidlin–Wentzell action $I[p] = \frac{1}{8}\int(\dot p - b)^2\,\mathrm{d}x$. Step 1 is to prove:

> *$\mathcal{J}$ is the linearization of Painlevé II about its Hastings–McLeod solution, in the spectral variable a; equivalently, the $\{c_n(a)\}$ are the PII trans-series coefficients. Then Costin's theorem gives Borel summability with singularities at the instanton action.*

2. The fluctuation operator (explicit)

Proposition 1 (Jacobi operator)**.** *Write $F[p] := \dot p - b = \dot p - (x+a) + p^2$, so $I = \frac{1}{8}\int F^2$. About the instanton (1), with $\eta := \delta p$,*

$$\delta^2 I[p_\star] = \tfrac{1}{4}\int \eta\,\mathcal{J}\,\eta\,\mathrm{d}x, \qquad \boxed{\mathcal{J} = -\partial_x^2 + Q(x), \quad Q(x) = 6p_\star^2 - 2(x+a).}$$

Proof. $\delta F = \dot\eta + 2p_\star\eta$ and $\delta^2 F = 2\eta^2$, so $\delta^2 I = \frac{1}{8}\int\big[2(\delta F)^2 + 2F_\star\,\delta^2 F\big] = \frac{1}{4}\int(\dot\eta + 2p_\star\eta)^2 + \frac{1}{2}\int F_\star\eta^2$, where on the instanton $F_\star = \dot{p_\star} - b_\star = -2b_\star$ (since $\dot{p_\star} = -b_\star$). Integrating $\int(\dot\eta + 2p_\star\eta)^2$ by parts, $\int\dot\eta^2 = -\int\eta\ddot\eta$ and $\int 4p_\star\eta\dot\eta = -2\int\dot{p_\star}\eta^2$, gives $\frac{1}{4}\int\eta\big[-\ddot\eta + (4p_\star^2 - 2\dot{p_\star})\eta\big] - \int b_\star\eta^2$. Using $\dot{p_\star} = p_\star^2 - (x+a)$ and $b_\star = (x+a) - p_\star^2$, $4p_\star^2 - 2\dot{p_\star} - 4b_\star = 6p_\star^2 - 2(x+a)$. ∎

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Corollary 2 (The Painlevé II fingerprint)**.** *The Painlevé II equation $q_{ss} = 2q^3 + sq$ linearizes about a solution q as $\delta_{ss} = (6q^2 + s)\delta$, i.e. the operator $-\partial_s^2 + (6q^2 + s)$. The fluctuation operator of Proposition 1 has* exactly *the PII coefficient $6(\cdot)^2$ on its field-squared term. A generic weak-noise Hessian would carry an arbitrary constant there; the value 6 is forced and is the isomonodromy fingerprint.* [computed here]

Remark 3 (What the two remaining discrepancies mean). $\mathcal{J} = -\partial_x^2 + 6p_\star^2 - 2(x + a)$ differs from $-\partial_s^2 + 6q^2 + s$ in (i) the sign/scale of the linear-in-position term and (ii) the field: $p_\star$ solves *Airy*, not PII. Both are resolved by the correct framing of §3: the linear term and the field belong to the *fixed-level* (x) problem, which is free (Airy); the PII lives in the deformation variable a, and $p_\star$ is the a-family's generating field, whose flow — not whose x-profile — is PII. Corollary 2 is the trace this leaves on the x-Hessian.

3. The correct framing: PII lives in the spectral variable a

Proposition 4 (Fixed level is free; the $\beta = 2$ anchor is exactly PII)**.** *(i) At fixed a the instanton problem is the Airy equation (1); its irregular singularity at $x = \infty$ has a-independent Stokes data (translation $z = x + a$ removes a). Hence the leading (Airy) deformation in a is trivial, and any PII structure is carried by the interacting fluctuations, i.e. by the $6p_\star^2$ term of $\mathcal{J}$.* [classical, cited]/[computed here] *(ii) At $\beta = 2$ the tail is* exactly *Painlevé II: by Tracy–Widom, $F_2(a) = \exp\big(-\int_a^\infty (x - a)\, q(x)^2\, \mathrm{d}x\big)$ with q the Hastings–McLeod solution of $q_{ss} = 2q^3 + sq$, and $\mathcal{T}_2(a) = 1 - F_2(a)$.* [classical, cited]

Thus a is the Painlevé (deformation) variable and x is the internal/spectral variable of the associated linear problem. The classical structure that governs this is:

- (FN) the Flaschka–Newell 2×2 linear system whose isomonodromic deformation is PII, with an irregular singularity carrying the Stokes/connection data; [classical, cited]
- (JMU) the Jimbo–Miwa–Ueno τ-function of that isomonodromic family, whose logarithmic derivative in the deformation variable is the Hamiltonian of PII. [classical, cited]

The Airy problem (1) is the degeneration of the FN linear system in which the PII field is switched off; turning on the fluctuations ($6p_\star^2$) deforms it, and the deformation in a is the PII flow.

4. Reduction of Step 1 to one τ-function identity

Claim 5 (Step 1 reduced)**.** *Let $\mathcal{D}(a;\beta) := \det_\zeta \mathcal{J}$ be the (zero-mode-removed, O4) fluctuation determinant of $\mathcal{J}$, and let $\tau_{\mathrm{JMU}}(a)$ be the Jimbo–Miwa τ-function of the a-isomonodromic PII family of §3. If*

$$\sum_{n \geq 0} c_n(a)\, \beta^{-n} \doteq \text{the large-}\beta \text{ asymptotic expansion of } \tau_{\mathrm{JMU}}(a) \text{ along the Hastings–McLeod trans-series,} \tag{2}$$

then the $c_n(a)$ are the PII trans-series coefficients, and Costin's theorem (rank-one nonlinear ODE at an irregular singularity) gives their Borel summability in $1/\beta$ with singularities at the instanton action, closing O1* and hence Theorem T1. [remaining]

Two independent supports that (2) is the right target, beyond Corollary 2:

- (S1) *Exact at $\beta = 2$.* Proposition 4(ii): there $\mathcal{T}_2(a)$ *is* the Hastings–McLeod expression, so (2) holds identically (not merely asymptotically) at $\beta = 2$, fixing the τ-function normalization the large-β expansion must match. [classical, cited]
- (S2) *Correct reality/Stokes structure.* The Hastings–McLeod trans-series has factorially divergent, non-sign-alternating coefficients in the relevant region, with the median-resummation reality condition (Cleri–Dunne); a linear transport hierarchy has no reason to reproduce that nonlinear Stokes structure, whereas a τ-function identification does. [classical, cited]

A concrete sub-computation toward (2). The a-derivative of the instanton, $\psi := \partial_a p_\star$, is a Jacobi field: differentiating (1) gives the linear inhomogeneous equation

$$\dot{\psi} = 2p_\star \psi - 1. \tag{3}$$

Its homogeneous part is annihilated by the same operator whose second-order form is $\mathcal{J}$; the inhomogeneity -1 is the source that, iterated through the transport hierarchy, generates the a-dependence of the c_n. The precise statement to establish is that the pair $(p_\star, \psi)$ solving (1)–(3) maps to the Flaschka–Newell frame of PII, so that $\partial_a \log \mathcal{D} = \partial_a \log \tau_{\mathrm{JMU}}$ up to the explicit one-loop term. This is a finite (Lax-pair) identification, the natural next write-up. [remaining]

5. Status

Computed here: the fluctuation operator $\mathcal{J} = -\partial_x^2 + 6p_\star^2 - 2(x+a)$ (Prop. 1) and its Painlevé II fingerprint (Cor. 2); the Jacobi-field equation (3). *Classical, cited (the backbone):* Airy triviality at fixed level and the Tracy–Widom $\beta = 2$ PII identity (Prop. 4); the Flaschka–Newell linear problem and the JMU τ-function. *Remaining (the crux of Step 1):* the τ-function identity (2), reduced to the finite Lax-pair identification of the sub-computation. Once it holds, Costin closes summability. *Assessment:* the fingerprint (Cor. 2) plus the exact $\beta = 2$ anchor (S1) make the PII identification very likely correct; Step 1 is now a bounded isomonodromy computation, not an open-ended search. Recommended next write-up: the Lax-pair map of (1)–(3) into the Flaschka–Newell frame.

References

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